Feedback and Partial Message Side-Information on the Semideterministic Broadcast Channel

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***Abstract*—The capacity of the semideterministic discrete memoryless broadcast channel (SD-BC) with partial message side-information (P-MSI) at the receivers is established. It is shown that P-MSI to the deterministic receiver can only increase capacity if also the stochastic receiver has P-MSI, while P-MSI to the stochastic receiver alone can increase capacity.**

These capacity results are used to show that on the SD-BC with or without P-MSI feedback from the stochastic receiver can increase capacity; with P-MSI at the deterministic receiver it can, in particular, increase the sum-rate capacity. When the stochastic receiver has full MSI (F-MSI), feedback cannot increase capacity.

I. Introduction

We establish the capacity region of the semideterministic discrete memoryless broadcast channel (SD-BC) when each receiver has partial message side-information (P-MSI), i.e., partial knowledge of the other receiver's message. Generally, the capacity region with P-MSI at the receivers is larger than without MSI. In particular, P-MSI at the stochastic receiver only can increase capacity on the SD-BC (see Example 1). The symmetric statement is not true: P-MSI at the deterministic receiver only cannot increase capacity (Corollary 5). Only when the stochastic receiver has P-MSI, P-MSI at the deterministic receiver can increase capacity (Note 4).

The capacity of the SD-BC without MSI was derived in [1], [2]. Kramer and Shamai [3] introduced and solved the BC with full-MSI (F-MSI) at both receivers, i.e., when each receiver knows the other receiver's message. They also obtained the capacity region of the BC with degraded message sets and P-MSI. P-MSI on the three-receivers BC is discussed in [4].

We use our results for the SD-BC with P-MSI (and a transmission scheme which—roughly speaking—turns feedback into P-MSI at the stochastic receiver) to show that feedback can increase the capacity of the SD-BC with or without P-MSI and, in particular, the sum-rate capacity when the deterministic receiver has P-MSI (Theorem 7). These results are contrasted by our last finding: on the SD-BC feedback cannot increase capacity when the stochastic receiver has F-MSI (Theorem 8).

To the best of our knowledge, our results are the first to show that feedback can increase the capacity of the SD-BC. While feedback does neither increase the capacity of the deterministic BC nor the capacity of the physically-degraded BC [5], it is known that feedback increases the capacity of several BCs with two stochastic receivers [6]–[9]. Achievable regions for the BC with feedback can be found in [7]–[10], but even with those at hand showing that feedback can increase the capacity of the SD-BC is nontrivial.

II. Channel Model

The SD-BC with finite input alphabet $\mathcal{X}$ and output alphabets $\mathcal{Y}$ and $\mathcal{Z}$ is characterized by a transition law of the form

$$W(y,z|x) = \mathbb{1}_{\{y=f(x)\}} W(z|x).$$

The encoder transmits an n-tuple X^n in order to convey two independent messages $M_{\mathcal{Y}}$ and $M_{\mathcal{Z}}$ to the deterministic receiver $\mathcal{Y}$ and the stochastic receiver $\mathcal{Z}$, respectively, where for each $\nu \in \{\mathcal{Y},\mathcal{Z}\}$ message M_ν is uniformly drawn from a set of size 2^{nR_ν}. We study the SD-BC with P-MSI, and we thus assume that each message comprises two parts, i.e.,

$$M_\nu = M_\nu^{(p)} \times M_\nu^{(c)},\ \nu \in \{\mathcal{Y},\mathcal{Z}\},$$

that Receiver $\mathcal{Y}$ knows $M_{\mathcal{Z}}^{(c)}$ and decodes $M_{\mathcal{Y}}$ from $(Y^n, M_{\mathcal{Z}}^{(c)})$, and that Receiver $\mathcal{Z}$ knows $M_{\mathcal{Y}}^{(c)}$ and decodes $M_{\mathcal{Z}}$ from $(Z^n, M_{\mathcal{Y}}^{(c)})$. For each $\nu \in \{\mathcal{Y},\mathcal{Z}\}$ we assume that $M_\nu^{(p)}$ and $M_\nu^{(c)}$ are independent of each other and uniformly drawn from sets of size $2^{nR_\nu^{(p)}}$ and $2^{nR_\nu^{(c)}}$, respectively, where $R_\nu = R_\nu^{(p)} + R_\nu^{(c)}$. Note the extreme cases:

- $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}) = (R_{\mathcal{Y}}, 0) \iff$ no MSI at Receiver $\mathcal{Z}$
- $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}) = (0, R_{\mathcal{Y}}) \iff$ F-MSI at Receiver $\mathcal{Z}$

Likewise for Receiver $\mathcal{Y}$. A rate-tuple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ is called achievable if there exists a sequence of encoders and decoders such that at each receiver the probability of a decoding error tends to zero as n tends to infinity. The capacity region is the closure of the set of all achievable rate-tuples.

We study the SD-BC with P-MSI in the absence and in the presence of feedback. In the absence of feedback, the encoder selects the channel input sequence as a function of the pair $(M_{\mathcal{Y}}, M_{\mathcal{Z}})$, i.e., $X^n = \phi(M_{\mathcal{Y}}, M_{\mathcal{Z}})$. This setting corresponds to that of Figure 1 without the dashed links.

When there is feedback, it is assumed to be one-sided from the stochastic receiver $\mathcal{Z}$ only. (Feedback from the deterministic receiver $\mathcal{Y}$ is useless because the transmitter can always compute Y^n from X^n.) We consider both perfect and rate-limited feedback. Perfect feedback allows the encoder to form the time-i input also as a function of Z^{i-1}, i.e., $X_i = \phi_i(M_{\mathcal{Y}}, M_{\mathcal{Z}}, Z^{i-1})$, $i \in [1:n]$. Rate-limited feedback of rate R_{FB} allows Receiver $\mathcal{Z}$ to transmit after Transmission i a signal $V_i = h(Z^i, M_{\mathcal{Y}}^{(c)}) \in \mathcal{V}_i$ to the encoder, and the encoder can in turn form the time-i input also as a function of V^{i-1}, i.e., $X_i = \phi_i(M_{\mathcal{Y}}, M_{\mathcal{Z}}, V^{i-1})$, $i \in [1:n]$. The rate-limitation implies that $\prod_{i=1}^n |\mathcal{V}_i| \le 2^{nR_{\text{FB}}}$. The SD-BC with P-MSI and perfect feedback (rate-limited feedback) corresponds

to the setting of Figure 1 where the dashed links transport the chance variables Z_i and Z^{i-1} (V_i and V^{i-1}). Note that perfect feedback is more powerful than rate-limited feedback: any rate-tuple that is achievable with rate-limited feedback can also be achieved with perfect feedback. Rate-limited and perfect feedback are equally powerful when $R_{\text{FB}} \geq \log |\mathcal{Z}|$.

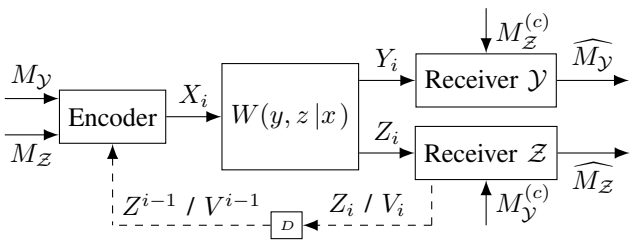

Fig. 1. SD-BC with P-MSI and feedback.

III. The SD-BC with P-MSI

In the absence of feedback, our main result is a single-letter characterization of the capacity of the SD-BC with P-MSI:

Theorem 1: The capacity region $\mathscr{C}_{\text{P-MSI}}$ of the SD-BC with P-MSI is the set of rate-tuples $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying

$$R_{\mathcal{Y}} \leq H(Y) \tag{1a}$$
$$R_{\mathcal{Z}} \leq I(U;Z) \tag{1b}$$
$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \leq I(V;Y) + H(Y|U) + I(U;Z|V) \tag{1c}$$
$$R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}} \leq H(Y|U) + I(U;Z) \tag{1d}$$
$$R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq I(V;Y) + H(Y|U) + I(U;Z) \tag{1e}$$

for some PMF of the form

$$p(v,u,x,y,z) = p(v,u)p(x|u)W(y,z|x). \tag{2}$$

W.l.g., one can restrict X to be a function of (U,Y).

To prove the direct part, we use a variation on Marton's code (see [11, Section 8.4]) for a common message $(M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)})$ and private messages $M_{\mathcal{Y}}^{(p)}$ and $M_{\mathcal{Z}}^{(p)}$. The random variable V represents the cloud center, and Y and U represent the satellites associated with $M_{\mathcal{Y}}^{(p)}$ and $M_{\mathcal{Z}}^{(p)}$, respectively. In their decoding, Receiver $\mathcal{Y}$ turns to account its knowledge of $M_{\mathcal{Z}}^{(c)}$ and likewise Receiver $\mathcal{Z}$ its knowledge of $M_{\mathcal{Y}}^{(c)}$.

Without any MSI, i.e., when $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$ and $R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$, Constraints (1c) and (1e) are redundant, and we recover:

Corollary 2 (From [1], [2], see also Section 8.3.1 of [11]): The capacity region $\mathscr{C}$ of the SD-BC without MSI is the set of rate-pairs $(R_{\mathcal{Y}}, R_{\mathcal{Z}})$ satisfying

$$R_{\mathcal{Y}} \leq H(Y) \tag{3a}$$
$$R_{\mathcal{Z}} \leq I(U;Z) \tag{3b}$$
$$R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq H(Y|U) + I(U;Z) \tag{3c}$$

for some PMF of the form

$$p(u,x,y,z) = p(u,x)W(y,z|x). \tag{4}$$

With F-MSI at the stochastic receiver $\mathcal{Z}$, Theorem 1 gives:

Corollary 3: For each $R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$, the capacity region of the SD-BC with F-MSI at Receiver $\mathcal{Z}$ ($R_{\mathcal{Y}}^{(p)} = 0$) is the set of rate-tuples $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying

$$R_{\mathcal{Y}} \leq H(Y) \tag{5a}$$
$$R_{\mathcal{Z}} \leq I(X;Z) \tag{5b}$$
$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \leq I(X;Y,Z) \tag{5c}$$

for some PMF of the form

$$p(x,y,z) = p(x)W(y,z|x). \tag{6}$$

Note 4: On the SD-BC, P-MSI to the stochastic receiver only can increase capacity, and when the stochastic receiver has P-MSI, then P-MSI to the deterministic receiver can also increase capacity.

The following example proves this note:

Example 1: Consider the SD-BC with binary input X, state $S \sim \text{Ber}(p)$, $p \in (0, 1/2)$, and outputs $Y = X$ and $Z = X \oplus S$. The capacity region $\mathscr{C}$ without MSI is the set of rate-pairs $(R_{\mathcal{Y}}, R_{\mathcal{Z}})$ that satisfy for some $\alpha \in [0, 1/2]$ (see [11, Section 5.4.2])

$$R_{\mathcal{Y}} \leq h_{\text{b}}(\alpha) \tag{7a}$$
$$R_{\mathcal{Z}} \leq 1 - h_{\text{b}}(\alpha + p - 2\alpha p). \tag{7b}$$

In particular, the sum-rate is bounded away from 1 unless $R_{\mathcal{Z}} = 0$. By Corollary 3 (with $X \sim \text{Ber}(1/2)$), the capacity region $\mathscr{C}_{\text{P-MSI}}$ with F-MSI at the stochastic receiver $\mathcal{Z}$ ($R_{\mathcal{Y}}^{(p)} = 0$) is the set of rate-tuples $(0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying

$$R_{\mathcal{Y}} \leq 1 \tag{8a}$$
$$R_{\mathcal{Z}} \leq 1 - h_{\text{b}}(p) \tag{8b}$$
$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \leq 1. \tag{8c}$$

When the deterministic receiver $\mathcal{Y}$ has no MSI ($R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Y}}$) and the stochastic receiver $\mathcal{Z}$ has F-MSI ($R_{\mathcal{Y}}^{(p)} = 0$), the sum-rate can thus be arbitrarily close to 1 irrespective of $R_{\mathcal{Z}}$.

In contrast, P-MSI at the deterministic receiver $\mathcal{Y}$ only cannot increase capacity:

Corollary 5: When the stochastic receiver $\mathcal{Z}$ has no MSI ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$), then irrespective of $R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$

$$(R_{\mathcal{Y}}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}} \iff (R_{\mathcal{Y}}, R_{\mathcal{Z}}) \in \mathscr{C}. \tag{9}$$

Proof: This follows from (1a), (1b), and (1d). ■

A. Intuition on the Results

On the two-user BC with independent messages, the transmission to each of the two receivers is interfered by the transmission to the other receiver. Since the transmitter knows the two messages and thus (non-causally) the two transmissions, it can partially mitigate the interference experienced at each receiver (see Marton's scheme [1]).

When one of the two receivers has P-MSI, it has knowledge about the transmission to the other receiver. In general, knowing interference both at the transmitter and the receiver

is more beneficial (in terms of achievable rates) than knowing it only at the transmitter (cf. the single-user channel with non-causal state information). As a consequence, with P-MSI a receiver can better mitigate the interference caused to its intended transmission than without MSI, and thus P-MSI at only one receiver typically increases capacity. An exception are deterministic channels: if the channel output is a function of some interference (e.g. a state sequence) and the channel input, then knowing interference non-causally only at the transmitter is as beneficial as knowing it also at the receiver. This may provide intuition on why P-MSI at the deterministic receiver only cannot increase the capacity, but P-MSI at the stochastic receiver only can.

Interference can be mitigated more efficiently when both receivers have P-MSI. For example, if $R_{\mathcal{Y}}^{(c)} = R_{\mathcal{Z}}^{(c)}$, then the transmitter can send the 'x-or' of the messages $M_{\mathcal{Y}}^{(c)}$ and $M_{\mathcal{Z}}^{(c)}$ as a common message to both receivers.

IV. Feedback on the SD-BC

This section investigates how feedback can be used to further mitigate interference on the SD-BC. We consider the cases with and without P-MSI at the receivers.

In the absence of MSI, (even one-sided) feedback can increase the capacity of the BC if both receivers are stochastic [6]. A common intuition for this is that feedback allows the transmitter to create a message that is beneficial to both receivers and, in doing so, to alleviate the interference at each receiver. In previously-presented examples, the transmitter typically creates a common message comprising information about past noise at the receivers, most prominently in Dueck's example [6].

The SD-BC has only one noisy receiver, and for the deterministic receiver learning the noise is useless. Thus, to prove that feedback can increase capacity on the SD-BC, a new intuition is required. In the following section, we present a feedback coding scheme that can increase capacity on the SD-BC with and without P-MSI (see Theorem 7).

A. Sketch of a Coding Scheme with Rate-Limited Feedback

For simplicity of exposition, we assume that only the deterministic receiver $\mathcal{Y}$ has P-MSI but not the stochastic receiver $\mathcal{Z}$, i.e., $R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$ and $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$. We partition the blocklength-n transmission into two phases: Phase 1 of length αn and Phase 2 of length $(1-\alpha)n$. In Phase 1 the encoder codes for an *enhanced BC* (with P-MSI at the deterministic receiver $\mathcal{Y}$) that is obtained from the original SD-BC by revealing the outputs observed at the stochastic receiver $\mathcal{Z}$ also to the deterministic receiver $\mathcal{Y}$. As we explain next, we use the feedback to convey Receiver $\mathcal{Z}$'s phase-1 channel outputs to Receiver $\mathcal{Y}$, allowing the latter to decode its phase-1 message. After Phase 1, the stochastic receiver $\mathcal{Z}$ compresses its phase-1 channel outputs and sends the compression index over the rate-limited feedback-link to the transmitter, which then conveys it to the deterministic receiver $\mathcal{Y}$ in the subsequent Phase 2.[1] Simultaneously, the transmitter sends fresh message information to each receiver. Since the stochastic receiver $\mathcal{Z}$ knows the compression index, it can use it as P-MSI in Phase 2. Hence, in Phase 2 the encoder codes for the SD-BC with P-MSI at *both* receivers.

At the end of each phase, the stochastic receiver $\mathcal{Z}$ decodes the message sent to it during that phase. The deterministic receiver $\mathcal{Y}$ waits with decoding until the end of Phase 2, when it first decodes its intended phase-2 message and the compression index. From the compression index it estimates Receiver $\mathcal{Z}$'s phase-1 channel outputs. It then decodes its intended phase-1 message based on its phase-1 channel outputs and its estimate of Receiver $\mathcal{Z}$'s phase-1 channel outputs.

The described scheme achieves the rate-region in Proposition 6 ahead, which depends on the capacity region $\mathscr{C}_{\text{enh}}$ of the enhanced BC without MSI.[2] Recall that $\mathscr{C}_{\text{enh}}$ is the set of rate-pairs $(R_{\mathcal{Y}}, R_{\mathcal{Z}})$ satisfying

$$R_{\mathcal{Y}} \leq I(X; Y, Z|U) \tag{10a}$$

$$R_{\mathcal{Z}} \leq I(U; Z) \tag{10b}$$

for some PMF of the form (4).

Proposition 6: Fix $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}$ and $(\tilde{R}_{\mathcal{Y}}, \tilde{R}_{\mathcal{Z}}) \in \mathscr{C}_{\text{enh}}$. If there exist a PMF $p(u, x, y, z)$ of the form (4) and a positive real number α such that the rate-pair $(\tilde{R}_{\mathcal{Y}}, \tilde{R}_{\mathcal{Z}})$ satisfies (10) and such that

$$\alpha \leq \min\left\{ \frac{R_{\mathcal{Y}}^{(c)}}{R_{\mathcal{Y}}^{(c)} + H(Z|Y)}, \frac{R_{\text{FB}}}{H(Z|Y)} \right\}, \tag{11}$$

then for any $\tilde{R}_{\mathcal{Z}}^{(p)}$, $\tilde{R}_{\mathcal{Z}}^{(c)} \geq 0$ that sum up to $\tilde{R}_{\mathcal{Z}} = \tilde{R}_{\mathcal{Z}}^{(p)} + \tilde{R}_{\mathcal{Z}}^{(c)}$, the capacity region of the SD-BC with P-MSI at Receiver $\mathcal{Y}$ only and feedback of rate R_{FB} contains the rate-tuple

$$(1-\alpha)(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) + \alpha(\tilde{R}_{\mathcal{Y}}, 0, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}). \tag{12}$$

B. Main Results

Whether feedback can increase the capacity of the SD-BC with P-MSI depends on the nature of the P-MSI:

Theorem 7: When the stochastic receiver $\mathcal{Z}$ has no MSI ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$), feedback can increase the capacity of the SD-BC irrespective of whether or not the deterministic receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$). It can in particular increase the sum-rate capacity when Receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$).

Proof: See Example 3 in Section IV-D. ■

Theorem 8: When the stochastic receiver $\mathcal{Z}$ has F-MSI ($R_{\mathcal{Y}}^{(p)} = 0$), even perfect feedback cannot increase the capacity

[1] The stochastic receiver $\mathcal{Z}$ uses the feedback-link only between the two phases. Therefore, we choose the alphabet $\mathcal{V}_i$ of the feedback V_i (see Figure 1) to be of size 1 if $i \neq \alpha n$ and of size $2^{nR_{\text{FB}}}$ if $i = \alpha n$. Of course, one could use the feedback-link more evenly and make sure that each alphabet $\mathcal{V}_i$ is of size at most $2^{R_{\text{FB}}}$ by using B instances of the two phases, where $B \to \infty$, and by starting the transmission with sufficiently many instances of Phase 1.

[2] For ease of notation, we define $\mathscr{C}_{\text{enh}} \in (\mathbb{R}_0^+)^2$ as the capacity region of the enhanced BC without MSI, even though in our coding scheme we code for the enhanced BC with P-MSI at Receiver $\mathcal{Y}$. Note that on the enhanced BC P-MSI at Receiver $\mathcal{Y}$ only cannot increase capacity, because if Receiver $\mathcal{Z}$ can decode $M_{\mathcal{Z}}$ then so can Receiver $\mathcal{Y}$, and thus Receiver $\mathcal{Y}$ can always generate F-MSI itself.

of the SD-BC. This holds irrespective of whether or not Receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$).

Proof: We prove that the rate-tuples that satisfy (5) for some PMF of the form (6) outer-bound the feedback capacity region. Let $Q \sim \text{Unif}[1:n]$ be independent of $(M_{\mathcal{Y}}, M_{\mathcal{Z}})$ and denote $(X, Y, Z) \triangleq (X_Q, Y_Q, Z_Q)$. We have

$$R_{\mathcal{Y}} - \epsilon_n \overset{(a)}{\leq} \frac{1}{n} I\big(M_{\mathcal{Y}}; Y^n, M_{\mathcal{Z}}^{(c)}\big) \tag{13}$$

$$\overset{(b)}{\leq} H(Y), \tag{14}$$

where (a) follows from Fano's inequality; and where (b) is true because of the chain-rule, because $M_{\mathcal{Y}}$ and $M_{\mathcal{Z}}^{(c)}$ are independent, because conditional entropy is nonnegative, and because conditioning cannot increase entropy. Similarly,

$$R_{\mathcal{Z}} - \epsilon_n \leq \frac{1}{n} I(M_{\mathcal{Z}}; Z^n, M_{\mathcal{Y}}) \tag{15}$$

$$\leq I(M_{\mathcal{Z}}; Z_Q | M_{\mathcal{Y}}, Z^{Q-1}, Q) \tag{16}$$

$$\overset{(a)}{\leq} I(X; Z), \tag{17}$$

where (a) holds since conditioning cannot increase entropy and since Z, X, and $(M_{\mathcal{Y}}, M_{\mathcal{Z}}, Z^{Q-1}, Q)$ form a Markov chain in that order. Note that if the probability of a decoding error is small, then $M_{\mathcal{Y}}$ is with high probability computable from $(Y^n, M_{\mathcal{Z}}^{(c)})$ and $M_{\mathcal{Z}}^{(p)}$ from $(Z^n, M_{\mathcal{Y}})$ and thus from $(Y^n, Z^n, M_{\mathcal{Z}}^{(c)})$. Hence, the probability of a decoding error is small only if $(M_{\mathcal{Y}}, M_{\mathcal{Z}}^{(p)})$ is with high probability computable from $(Y^n, Z^n, M_{\mathcal{Z}}^{(c)})$, and thus we conclude that

$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(c)} - \epsilon_n$$

$$\overset{(a)}{\leq} \frac{1}{n} I\big(M_{\mathcal{Y}}, M_{\mathcal{Z}}^{(p)}; Y^n, Z^n, M_{\mathcal{Z}}^{(c)}\big) \tag{18}$$

$$= I\big(M_{\mathcal{Y}}, M_{\mathcal{Z}}^{(p)}; Y_Q, Z_Q \big| M_{\mathcal{Z}}^{(c)}, Y^{Q-1}, Z^{Q-1}, Q\big) \tag{19}$$

$$\leq I(X; Y, Z), \tag{20}$$

where (a) follows from Fano's inequality. ■

Of course, whether feedback can increase the capacity of the SD-BC also depends on the channel. The following is an example of an SD-BC that is neither deterministic nor physically-degraded, but still feedback does not increase its capacity without MSI at Receiver $\mathcal{Z}$, irrespective of whether or not Receiver $\mathcal{Y}$ has P-MSI.

Example 2: Consider the SD-BC with binary state $S \sim \text{Ber}(p)$, $p \in (0,1)$, and outputs

$$Y = f(X), \quad Z = \begin{cases} X, & S = 0, \\ ?, & S = 1, \end{cases} \tag{21}$$

where $\mathcal{Y} \subsetneq \mathcal{X}$, and $? \notin \mathcal{X}$. It is easy to see that for the considered SD-BC the capacity region $\mathscr{C}$ without MSI is the set rate-pairs satisfying

$$R_{\mathcal{Y}} \leq H(Y) \tag{22a}$$

$$R_{\mathcal{Z}} \leq \bar{p} I(U; Y) + \bar{p} \sum_{y \in \mathcal{Y}} p(y) \log |\mathcal{X}_y| \tag{22b}$$

$$R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq H(Y) - p I(U; Y) + \bar{p} \sum_{y \in \mathcal{Y}} p(y) \log |\mathcal{X}_y| \tag{22c}$$

for some PMF $p(u, x, y, z)$ of the form (4), where for every $y \in \mathcal{Y}$ we denote $\mathcal{X}_y \triangleq \{x \in \mathcal{X} \colon f(x) = y\}$. Moreover, it is not difficult to show that the feedback capacity region of any SD-BC without MSI at Receiver $\mathcal{Z}$ ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$) is outer-bounded by the set of all rate-tuples satisfying (3a), (3b), and

$$R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq I(X; Y, Z | U) + I(U; Z) \tag{23}$$

for some PMF $p(u, x, y, z)$ of the form (4), irrespective of whether or not Receiver $\mathcal{Y}$ has MSI ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$). For the considered BC, this region coincides with $\mathscr{C}$.

Similar results as in Theorems 7 and 8 hold if the SD-BC is state-dependent and the encoder is furnished with strictly-causal state information (SI) instead of feedback:

Note 9: Suppose that the SD-BC is governed by an i.i.d. state-sequence of finite support $\mathcal{S}$ in the sense that

$$W(y, z | x, s) = \mathbb{1}_{\{y = f(x)\}} W(z | x, s), \; s \in \mathcal{S}.$$

When the stochastic receiver $\mathcal{Z}$ has no MSI ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$), strictly-causal SI can increase the capacity of the SD-BC irrespective of whether or not the deterministic receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$), and it can in particular increase the sum-rate capacity when Receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$). When the stochastic receiver $\mathcal{Z}$ has F-MSI ($R_{\mathcal{Y}}^{(p)} = 0$), strictly-causal SI cannot increase capacity irrespective of whether or not Receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$).

C. Intuition on the Results with Feedback

Based on our coding scheme in Section IV-A, we can offer the following intuition: feedback on the SD-BC (with or without P-MSI) can increase capacity because it allows the encoder to build up P-MSI at the stochastic receiver and thus allows the stochastic receiver to better mitigate experienced interference (see Note 4 and the discussion in Section III-A). The setting where the stochastic receiver has F-MSI is different: feedback cannot increase capacity because F-MSI suffices for the stochastic receiver to fully mitigate experienced interference.

D. On the Proof of Theorem 7

We prove Theorem 7 by means of the following example:

Example 3: On the SD-BC with binary inputs X_1 and X_2, binary state $S \sim \text{Ber}(p)$, $p \in (0,1)$, and outputs

$$Y = X_1 + X_2, \quad Z = \begin{cases} X_2, & S = 0, \\ ?, & S = 1, \end{cases} \tag{24}$$

rate-limited feedback increases the sum-rate capacity when Receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$) and Receiver $\mathcal{Z}$ has no MSI ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$). It increases the capacity region when $p > 1/2$ and no receiver has MSI ($R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$ and $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$).

The analysis of the example is based on the following two theorems that identify conditions when feedback increases

capacity of the SD-BC. Theorem 10 treats the case with P-MSI at Receiver $\mathcal{Y}$ and Theorem 11 the case without MSI.

Theorem 10: Consider an SD-BC where only the deterministic receiver $\mathcal{Y}$ has MSI ($R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$ and $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$). If there exists a rate-tuple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying[3,4]

$$(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)} + R_{\mathcal{Z}}^{(c)}) \in \big(\partial\mathscr{C} \cap (\mathscr{C}_{\text{enh}} \setminus \partial\mathscr{C}_{\text{enh}})\big), \qquad (25)$$

and if for some PMF $p(u, x, y, z)$ of the form (4) Conditions (3) and

$$R_{\mathcal{Y}} < H(Y) \quad \text{and} \quad 0 < I(U; Y) \qquad (26)$$

hold, then $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ is in the interior of the feedback capacity region, and feedback thus increases capacity.

Theorem 11: Consider an SD-BC without MSI ($R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$ and $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$). If there exists a rate-pair $(R_{\mathcal{Y}}, R_{\mathcal{Z}}) \in \big(\partial\mathscr{C} \cap (\mathscr{C}_{\text{enh}} \setminus \partial\mathscr{C}_{\text{enh}})\big)$ such that for some PMF $p(u, v, x, y, z)$ of the form (2) Conditions (3) and

$$R_{\mathcal{Y}} < H(Y) \quad \text{and} \quad 0 < I(V; Y) - I(V; Z) \qquad (27)$$

hold, then $(R_{\mathcal{Y}}, R_{\mathcal{Z}})$ is in the interior of the feedback capacity region, and feedback thus increases capacity.

In the following, we sketch the proof of Theorem 10. A first step is to show that, under the conditions stated in the theorem, it is possible to identify a rate-triple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ that satisfies the following two conditions:

1) $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)} + R_{\mathcal{Z}}^{(c)})$ lies on the boundary of the no-feedback capacity region $\mathscr{C}$ and in the interior of the capacity region $\mathscr{C}_{\text{enh}}$ of the enhanced BC.
2) There exists a strictly-positive rate $R_{\mathcal{Y}}^{(c)}$ such that $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ is contained in $\mathscr{C}_{\text{P-MSI}}$.

As we argue next, the above conditions guarantee that the feedback coding scheme of Section IV-A achieves a rate-tuple outside the no-feedback capacity region $\mathscr{C}$. (Recall from Corollary 5 that $\mathscr{C}$ is also the capacity of the SD-BC with P-MSI at Receiver $\mathcal{Y}$ only.) Indeed, Condition 1 guarantees that in Phase 1 of the feedback code, when the encoder codes for the enhanced BC, message information can be sent at rates $(\tilde{R}_{\mathcal{Y}}, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)})$ satisfying

$$\tilde{R}_{\mathcal{Y}} > R_{\mathcal{Y}}^{(p)}, \quad \tilde{R}_{\mathcal{Z}}^{(p)} > R_{\mathcal{Z}}^{(p)}, \quad \tilde{R}_{\mathcal{Z}}^{(c)} > R_{\mathcal{Z}}^{(c)}. \qquad (28)$$

Recall that the deterministic receiver $\mathcal{Y}$ needs resolution information to decode its phase-1 message. Condition 2 ensures that in Phase 2 of the feedback code the encoder can transmit the required resolution information at rate $R_{\mathcal{Y}}^{(c)}$ and simultaneously send fresh message information at rates $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$. Specifically, Proposition 6 implies that for α sufficiently small the rate-tuple

$$(1-\alpha)(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) + \alpha(\tilde{R}_{\mathcal{Y}}, 0, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}). \qquad (29)$$

is achievable. Since $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)} + R_{\mathcal{Z}}^{(c)})$ lies on the boundary of the no-feedback capacity region $\mathscr{C}$, we conclude from (28) that the rate-tuple (29) is not achievable without feedback, and that feedback thus increases capacity.

The only difference in the proof of Theorem 11 is that we set $R_{\mathcal{Z}}^{(c)} = \tilde{R}_{\mathcal{Z}}^{(c)} = 0$.

Note 12: Theorem 11 also follows from [9, Cor. 3].[5]

To prove that feedback can increase the capacity region of the SD-BC without MSI, we invoke Theorem 11 with the choice $V = U$. This raises the question if choosing $V \neq U$ can help. As we argue next, this is indeed the case. To see this, extend the SD-BC of Example 3 by a parallel channel of capacity larger than $\log 3$, and assume that Receiver $\mathcal{Z}$ additionally observes the output of the parallel channel. (E.g., assume that, in addition to Z, Receiver $\mathcal{Z}$ noiselessly observes an input X_3, which takes value in a set of size four.) For the constructed SD-BC, it is easy to see that a rate-tuple satisfying (3) for some PMF of the form (4) is a boundary point of the capacity region only if the auxiliary random variable U comprises a capacity-achieving input to the parallel channel. (E.g., if Receiver $\mathcal{Z}$ observes the pair (Z, X_3), then X_3 must have a uniform prior and be a function of U.) But this implies that $I(U; Y) \leq \log 3 < I(U; Z)$. Hence, we cannot invoke Theorem 11 with the choice $V = U$ to show that feedback can increase capacity. However the following choice of V does the trick: let V be the random variable that is obtained by discarding the input to the parallel channel from U.

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[3] We denote the boundary of any set $\mathscr{A}$ by $\partial\mathscr{A}$.

[4] Condition (25) implies that $(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \partial\mathscr{C}_{\text{P-MSI}}$ (cf. (9)).

[5] This may be surprising: [9, Cor. 3] says that feedback from Receiver 1 increases the capacity region of the BC $W(y_1, y_2|x)$ if for some PMF $p(u_0, u_1, u_2, x, y, z)$ for which $0 < I(U_0; Y_2) - I(U_0; Y_1)$ Marton's inner bound yields a rate-point (R_1, R_2) with $R_2 < I(U_0, U_2; Y_2)$ that is on the boundary of the BC's capacity region and in the interior of the capacity region of the enhanced BC, which is obtained from the original BC by revealing Y_1 also to Receiver 2. But for the SD-BC Marton's inner bound is tight if $U_0 = \emptyset$, and in this case we have $I(U_0; Y) = I(U_0; Z) = 0$.